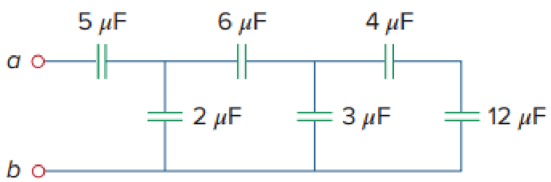
Chapter 6, Problem 21P

Problem

Determine the equivalent capacitance at terminals *a*-*b* of the circuit in Fig. 6.55.

Fig. 6.55:



Step-by-step solution

Step 1 of 1

$4\mu F$ in series with $12\mu F = (4\times 12)/16 = 3\mu F$

$3\mu F$ in parallel with $3\mu F = 6\mu F$

$6\mu F$ in series with $6\mu F = 3\mu F$

$3\mu F$ in parallel with $2\mu F = 5\mu F$

$5\mu F$ in series with $5\mu F = 2.5\mu F$.

Hence $C_{eq} = 2.5\,\mu F$